TECHNICAL REFERENCE

Developer Documentation

Tuntaskilat

Architecture, Data Model, Cloud Functions API, Security Rules, and Business Logic — a complete reference for the engineering team

PT Tuntas Kilat Group

Internal Engineering Reference

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Tuntaskilat
Developer Reference

On-demand home cleaning service platform — 3 Flutter client apps (Customer, Worker, Admin) on a shared Firebase backend. This reference covers the data model, Cloud Functions API, security rules, and business logic that power it.

Get Started → End-user Manual

3	22	20+	1
Client Apps	Cloud Functions	Firestore Collections	Shared Package (tk_core)

Explore the docs

Getting Started
Clone the repo, install dependencies, run each app locally.

Data Model
Full Firestore schema — every collection, field, and enum.

Cloud Functions

All 22 server-side functions: triggers, inputs, outputs.

Security Rules

Read/write access matrix for every collection.

AI Features

Multi-provider AI customer service & business analyst.

Payments

Static gateway + Xendit dynamic invoicing/webhook.

Tech Stack

stack.yaml

```yaml
client:
  framework: Flutter 3.41
  state: Riverpod 2.6
  language: Dart 3.11
  apps: [pelanggan, kru, admin]
  shared_package: packages/tk_core

backend:
  platform: Firebase
  functions: Node.js 20 (2nd Gen), TypeScript
  region: asia-southeast2
  database: Cloud Firestore (NoSQL, document-based)
  auth: Firebase Authentication (email/password + Google)
  storage: Cloud Storage
  messaging: Firebase Cloud Messaging (FCM)

integrations:
  ai: [Gemini, OpenAI, Anthropic]  # multi-provider, admin-configurable
  payments: [transfer_bank, qris, tunai, xendit]
```

```
  whatsapp: Fonnte
```

INTRODUCTION

Getting Started

Clone the monorepo, install dependencies, and get all three apps running locally.

Prerequisites

You'll need the following tools installed:

```
Flutter SDK    3.41.x (stable channel)
Dart SDK       3.11.x
Node.js        20.x
Firebase CLI   latest  (npm i -g firebase-tools)
A Firebase project with Firestore, Auth, Storage, Functions, FCM enabled
```

Repository Structure

This is a Flutter pub workspace monorepo — one Git repository, multiple apps sharing a common package:

```
tkapps/
├── apps/
│   ├── pelanggan/     # Customer app (Android)
│   ├── kru/           # Worker app (Android)
│   └── admin/         # Admin panel (Flutter Web)
├── packages/
│   └── tk_core/       # Shared models, services, theme, widgets
├── firebase/
```

```
│   ├── functions/       # Cloud Functions (TypeScript)
│   ├── firestore.rules
│   └── storage.rules
├── docs-site/
│   ├── manualbook/      # This site's sibling — end-user manual
│   └── documentation/   # You are here
├── firebase.json        # Hosting targets, functions, rules
└── pubspec.yaml          # Workspace root
```

Local Setup

```
git clone https://github.com/nabhanyuzqi1/tuntaskilat.git
cd tuntaskilat

# Flutter workspace — installs deps for all 3 apps + tk_core at once
flutter pub get

# Cloud Functions
cd firebase/functions
npm install
cd ../..

# Connect to your own Firebase project (or use the existing one)
firebase use --add
```

ⓘ Each app needs its own `firebase_options.dart`, generated via `flutterfire configure` — run it once per app (`apps/pelanggan`, `apps/kru`, `apps/admin`).

Running Each App

```
# Customer app (Android emulator/device)
cd apps/pelanggan && flutter run

# Worker app
cd apps/kru && flutter run

# Admin panel (web)
cd apps/admin && flutter run -d chrome

# Cloud Functions emulator
cd firebase && firebase emulators:start --only firestore,functions
```

Testing

```
# Unit + widget tests (business logic, pricing, security scenarios)
flutter test packages/tk_core
flutter test apps/admin

# Static analysis across the workspace
flutter analyze packages/tk_core apps/admin apps/pelanggan apps/kru

# Cloud Functions type-check
cd firebase/functions && npx tsc -p .
```

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Architecture

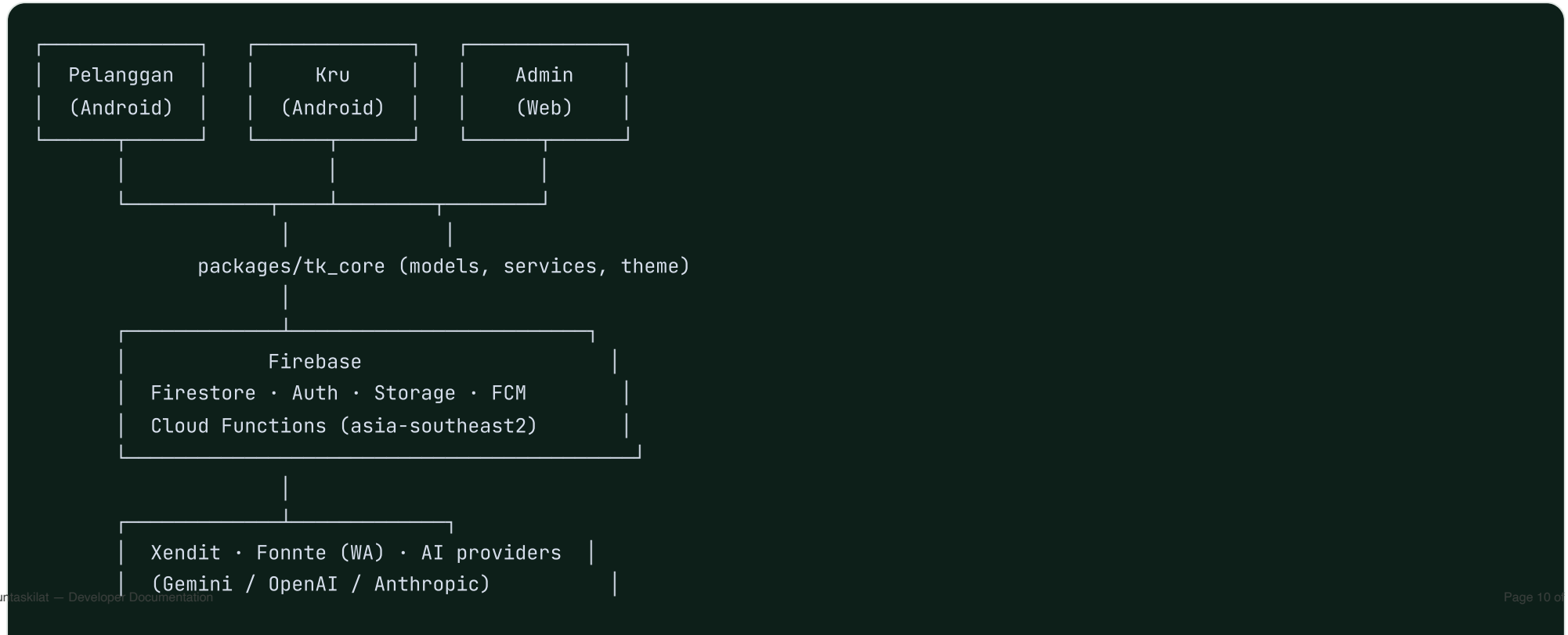
INTRODUCTION

Architecture

A monorepo of 3 Flutter clients sharing one package, backed by Firebase — Firestore as the source of truth, Cloud Functions as the money-critical safety net.

System Overview

Every client writes optimistically via a Firestore transaction (client-side pricing/locking), and every write of consequence is re-validated by a Cloud Function running with the Admin SDK — which bypasses Security Rules and is treated as the single source of truth for money. The client is a preview; the server is authoritative.



Client Applications

App	Platform	Users	Purpose
apps/pelanggan	Android	Customers	Browse services, book, pay, track, review, chat, AI CS.
apps/kru	Android	Field crew	Accept jobs, navigate, report work, manage cash deposits.
apps/admin	Flutter Web	Staff/owner	Manage orders, catalog, crew, vouchers, commission, operations.

Shared Package (tk_core)

`packages/tk_core` holds everything that must stay consistent across apps: Firestore models & enums, the `FirestoreService` facade (all reads/writes/transactions), auth flow, theme tokens (colors, typography, radius), and shared widgets. No app talks to Firestore directly — everything routes through `tk_core`.

Backend Services

Service	Role
Cloud Firestore	Document-based NoSQL database — the system of record for orders, users, payments, etc.
Firebase Auth	Email/password + Google sign-in. Role (pelanggan/kru/admin) stored on the users document, not in Auth claims.
Cloud Storage	Payment proof uploads, work-report photos, profile photos, service images.
Cloud Functions	22 functions (Node.js 20, TypeScript, region asia-southeast2) — triggers, scheduled jobs, and callables. See Functions reference.

Service	Role
FCM	Push notifications for order status changes, chat messages, assignment reminders, broadcasts.

Auth & Roles

A single `UserRole` enum (`pelanggan` | `kru` | `admin`) gates which app an account can log into. `AuthService.signIn()` checks the role against the app performing the login and signs the session out immediately on a mismatch (a pelanggan account cannot log into the Kru portal, etc.) — this check exists both client-side and mirrored in Security Rules.

Role	Self-registration?	How the account is created
pelanggan	Yes	registerPelanggan() — email/password or Google sign-in, self-service.
kru	No	Admin only, via A5 → AkunKruService.buatAkunKru() (uses a secondary Firebase App instance so the admin's own session is never disturbed).
admin	No	Existing admin only, via the buatAdmin callable Cloud Function (admin.auth().createUser() server-side).

ⓘ Admin accounts can additionally require TOTP 2FA (RFC 6238) at login — secret stored in `admin2fa/{uid}`, owner-only read/write.

Atomic Locking Pattern

Booking a time slot is a classic race condition — two customers could both read "slot open" and both write a booking. Tuntaskilat solves this without a dedicated locking service, using a Firestore transaction against a capacity counter:

simplified booking transaction

```
await db.runTransaction(async (tx) ⇒ {
```

```
  const slotSnap = await tx.get(slotRef)
  const { terisi = 0, kapasitas } = slotSnap.data() ?? {}

  if (terisi ≥ kapasitas) throw new JadwalPenuhException()

  // Firestore rule enforces this write can ONLY increment by exactly 1
  tx.set(slotRef, { terisi: terisi + 1, kapasitas }, { merge: true })
  tx.set(orderRef, order.toMap())
})
```

`kapasitas` (capacity) is not a fixed number — it's recomputed automatically whenever crew data changes (see `recomputeKapasitasLayanan()` in the Functions reference), so booking availability always reflects how many eligible crew are actually active for that service.

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Data Model

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Data Model

Every Firestore collection Tuntaskilat writes to — field names, types, and enum values, sourced directly from packages/tk_core/lib/models/.

Model source files live in `packages/tk_core/lib/models/` — one Dart class per collection, each with typed `fromMap()` / `toMap()` so every read/write goes through the same shape.

users/{userId}

`UserModel`

One document per account, docId = Firebase Auth UID. Shared shape across all 3 roles.

nama	`string`	
email	`string`	
noTelepon	`string`	
alamat	`string`	
role	`UserRole`	
fotoUrl	`string`	
nonaktif	`bool`	company-deactivated account flag
kodeReferal	`string`	deterministic 6-char base32 hash of uid, prefixed TK
referredBy	`string`	referral code entered at signup

Enums

UserRole	enum	pelanggan | kru | admin

Subcollection users/{uid}/alamat: { label, alamat, lokasi: GeoPoint } — saved addresses.

services/{serviceId}

`ServiceModel`

Catalog of bookable cleaning services with dynamic pricing.

namaLayanan	string	
deskripsi	string	
harga	num	base / "starting from" price
satuan	string	per jam | per ruangan | per m2
aktif	bool	
kategori	string	rumput | home_cleaning | umum
gambarUrl	string	
tipeHarga	TipeHarga	
tiers	TarifTier[]	{ id, nama, hargaPerM2 } — for per_luas pricing
paketOpsi	PaketOpsi[]	{ id, nama, jumlahPetugas, durasi[], hargaTambahJam, spesifikasi }
addOns	AddOn[]	{ id, nama, harga }
jumlahKru	num	capacity — maintained by Cloud Function, never written by admin edits

Enums

TipeHarga	enum	per_luas | paket | mulai_dari

orders/{orderId}

OrderModel

The central document — every transaction and its full lifecycle.

userId	string	
serviceId	string	
cleanerId	string	lead crew member, "" = unassigned
tanggalPesan / jadwal	Timestamp	
totalHarga / subtotal / potongan	num	totalHarga = subtotal − potongan
status	OrderStatus	
penugasan	Penugasan[]	{ cleanerId, nama, peran, sudahKonfirmasi, diterima, fotoUrl }
kruIds	string[]	for array-contains queries
metodePembayaran	MetodeBayar	
voucherKode / rincian	string / BarisRincian[]	
fotoSebelum / fotoSesudah	string[]	work-report photos
slotId	string	written manually, not in toMap() — links to the capacity slot
kasTunaiDibukukan	bool	idempotency guard for cash-commission booking

Enums

OrderStatus	enum	dibuat → menunggu_pembayaran → menunggu_verifikasi → (ditolak ↩ | terverifikasi) → menunggu_penugasan → ditugaskan → dalam_perjalanan → diproses → selesai → dinilai, + dibatalkan
MetodeBayar	enum	transfer_bank | qris | tunai

payments/{paymentId}

PaymentModel

Payment proof + verification status. Xendit adds ad-hoc gateway/invoiceId/invoiceUrl fields not in the Dart model.

orderId / userId	string	
metode	MetodeBayar	
jumlah	num	
buktiBayar	string?	Storage URL
statusBayar	StatusBayar	

Enums

StatusBayar	enum	menunggu | terverifikasi | ditolak

payouts/{orderId}_{cleanerId}

PayoutModel

Crew wage ledger — immutable to clients once created. Money is never trusted from the client after this point.

orderId / cleanerId / namaKru	string	

peran	`PeranKru`
jumlah	`num`
status	`StatusPayout`

Enums

PeranKru	`enum`	worker (weight 1.0) | helper (weight 0.6)
StatusPayout	`enum`	pending | dibayar | ditahan

kru/{cleanerId}

`KruModel`

Crew profile, docId = Firebase Auth UID.

nama / noTelepon / fotoUrl	`string`	
statusKetersediaan	`bool`	online/offline toggle
posisi	`GeoPoint?`	live location, streamed while dalam_perjalanan
rataRating / jumlahUlasan	`num`	server-authoritative, recomputed on every review
fcmTokens	`string[]`	
keahlian	`string[]`	serviceIds — empty array = generalist (eligible for all services)
tipe	`KruTipe`	
status	`StatusKru`	only aktif is assignable
rekening	`RekeningKru`	{ jenis: bank|ewallet, penyedia, nomor, atasNama }

Enums

KruTipe	enum	kru | mitra | vendor
StatusKru	enum	aktif | nonaktif | diberhentikan

kasKru/{cleanerId}

KasKru

Cash-deposit ledger — how much commission from cash-paid jobs a crew member is still holding.

namaKru	string	
saldoTunai	int	≥ 0. Invariant: saldoTunai == totalMasuk − totalSetor
batasNunggak	int	default 200000 (Rp) — arrears threshold
totalMasuk / totalSetor	int	lifetime accumulated in / deposited out

setoran/{setoranId}: { cleanerId, namaKru, jumlah, diterimaOleh, waktu, catatan } — immutable deposit history.

vouchers/{kode}

VoucherModel

docId = uppercase code.

tipe	TipeVoucher	
nilai	num	percent or fixed amount depending on tipe
minBelanja / maxPotongan	num	maxPotongan 0 = unbounded
kuota / terpakai	int	kuota 0 = unlimited
berlakuHingga	string?	ISO date

khususPenggunaBaru `bool`	
sekaliPerNomor `bool`	default true — locks claim per phone number, not per uid

Enums

TipeVoucher `enum`	persen | nominal

voucherUsages/{kode}__{telepon}: anti-abuse lock, immutable to the user once created.

reviews/{reviewId}

`ReviewModel`

Public — every review re-aggregates the rated crew member's kru.rataRating server-side (onReviewCreate).

orderId / userId / cleanerId `string`	
penilaian `num`	1–5
komentar / namaPelanggan `string`	

notifications/{id}

`NotificationModel`

In-app notification feed, mirrors FCM pushes.

userId `string`	recipient
judul / pesan `string`	
dibaca `bool`	
orderId `string?`	deep-link target

disputes/{disputeId}

`DisputeModel`

Appeals over work quality or payout amount.

orderId / pengajuId / pengajuNama	`string`
pengaju	`PengajuBanding`
status	`StatusBanding`
alasan / catatanAdmin	`string`
buktiUrl	`string?`

Enums

PengajuBanding	`enum`	kru | pelanggan
StatusBanding	`enum`	diajukan | diterima | ditolak

banners/{id} · broadcasts/{id}

`BannerModel · (none)`

Two unrelated concepts despite the similar name — see the AI/Notifications pages for how each is used.

banners.judul / subjudul / badge / gambarUrl / isi / tautan / urutan / aktif	– persistent, admin-curated promo content shown on the Beranda
broadcasts.judul / pesan / terkirim / tanpaToken / selesaiPada	– one-shot marketing push blast to all pelanggan, no dedicated model class

settings/{docId}

`various`

Runtime configuration, admin-controlled. See individual docIds below.

komisi	`KonfigKomisi`	global commission % + per-service overrides
pembayaran	`KonfigPembayaran`	which payment methods are active/static/dynamic
app	`KonfigApp`	maintenance mode / forced update — publicly readable
ai	`KonfigAi`	provider + API keys for AI features — admin-only, secret
referral	`{ nominal, minBelanja, masaBerlakuHari }`	referral reward configuration

referralClaims/{telepon}

`(none)`

One document per phone number, ever — the anti-abuse lock for the referral program.

telepon / buyerId / referrerId	`string`
kodeReferal / voucherKode	`string`

admin2fa/{uid}

`(none)`

TOTP secret for admin 2FA. Owner-only read/write.

secret	`string`
aktif	`bool`

slots/{slotId}

`(none)`

Capacity counter backing the atomic-locking booking mechanism.

serviceId / jadwal	`string / Timestamp`	
terisi / kapasitas	`num`	terisi can only be incremented by exactly +1 per client write (rule-enforced); decrements are admin-SDK only

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Cloud Functions

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Cloud Functions

22 server-side functions — all running in asia-southeast2 as a safety-net layer that mirrors and validates client-side transactions.

ⓘ Every function shares one Firestore admin instance and the `REGION = "asia-southeast2"` constant, defined in `firebase/functions/src/index.ts`.

Orders & Pricing Integrity

onOrderCreate `onDocumentCreated` orders/{orderId}

Re-reads the service document and recomputes the correct price server-side, correcting the order in place if the client-submitted totalHarga/subtotal/potongan drift by more than Rp1.

> Anti price-tampering: the client computes a preview price for UX, but this function is what actually decides what the customer owes.

validatePayment `onDocumentCreated` payments/{paymentId}

Confirms payment.jumlah equals the referenced order's totalHarga (tolerance Rp1); corrects it if mismatched.

onOrderFinalize `onDocumentUpdated` orders/{orderId}

Fires on transition into status: 'selesai'. Computes crew payouts (weighted split, helper ratio 0.6) and, for cash orders, books the platform commission into the lead crew member's kasKru ledger.

> Idempotent via the kasTunaiDibukukan flag on the order document — cash commission is only ever booked once.

onOrderAssigned `onDocumentUpdated` `orders/{orderId}`

Fires on transition into 'ditugaskan'. Sends FCM push to every assigned crew member's fcmTokens, cleaning up invalid tokens.

onOrderCancelled `onDocumentUpdated` `orders/{orderId}`

Fires on transition into 'dibatalkan'. Decrements the booking slot's terisi counter (or deletes the slot doc) — the only permitted decrement path, since clients can only increment.

pushStatusPelanggan `onDocumentUpdated` `orders/{orderId}`

Pushes FCM to the customer on every meaningful status transition, with data.tipe='status' for deep-linking into the tracking screen.

autoCancelTanpaKru `onSchedule` `every 60 minutes (Asia/Makassar)`

Cancels orders stuck in 'terverifikasi'/'menunggu_penugasan' for more than 24 hours with no crew assigned, notifying the customer.

Chat & Notifications

onChatMessageCreate `onDocumentCreated` `orders/{orderId}/messages/{messageId}`

Determines recipients (pelanggan → all kruIds; kru → userId), writes a notification doc, and pushes FCM.

reminderKru `onSchedule` `every 30 minutes (Asia/Makassar)`

Finds 'ditugaskan' orders whose jadwal is 90–150 minutes away and sends a reminder push to assigned crew.

onBroadcastCreate `onDocumentCreated` `broadcasts/{id}`

Mass-pushes an FCM notification to every user with role=='pelanggan', then writes delivery stats back onto the broadcast doc.

waNotifOrder `onDocumentUpdated` `orders/{orderId}`

Sends a WhatsApp message via Fonnte for terverifikasi/ditugaskan/selesai status transitions. No-op if FONNTE_TOKEN is unset.

Reviews & Capacity

onReviewCreate `onDocumentCreated` `reviews/{reviewId}`

Re-aggregates ALL reviews for the rated cleanerId and overwrites kru/{cleanerId}.rataRating / jumlahUlasan — server-authoritative, never client-computed.

onKruDitulis `onDocumentWritten` `kru/{cleanerId}`

Any create/update/delete of a crew doc triggers recomputeKapasitasLayanan().

onLayananDibuat `onDocumentCreated` `services/{serviceId}`

A new service triggers recomputeKapasitasLayanan() so jumlahKru is populated immediately.

backfillKapasitasKru `onCall` `admin-only`

Manually forces a full recompute of service capacities — used after bulk keahlian edits.

Admin & Referral

buatAdmin `onCall` `admin-only`

Input { email, password, nama }. Creates a Firebase Auth user + users doc (role: admin) without disturbing the caller's session. Returns { uid }.

setNonaktifAdmin `onCall` `admin-only`

Input { uid, nonaktif }. Disables/enables the target Auth account. Cannot target self (failed-precondition).

rewardReferral `onDocumentUpdated` `orders/{orderId}`

On a buyer's first completed order, mints a reward voucher for the referrer, guarded by a transaction on referralClaims/{phone} — one claim per phone number, ever.

Payments (Xendit)

xenditWebhook `onRequest` `HTTP endpoint`

Validates x-callback-token, then on status PAID/SETTLED marks the matching payment & order 'terverifikasi'. Returns 501 if XENDIT_CALLBACK_TOKEN is unset.

buatTagihanXendit `onCall` `auth required`

Input { orderId, amount }. Verifies order ownership, calls the Xendit Invoices API, persists invoiceUrl/invoiceId on the payment doc. Requires XENDIT_SECRET_KEY.

AI

csAi `onCall` `auth required`

Input { pertanyaan, orderId? }. Multi-provider chat (Gemini/OpenAI/Anthropic) grounded in the live service catalog + the caller's own order status. Returns { jawaban }.

analisaBisnisAi `onCall` `admin-only`

Aggregates the trailing 30 days of orders (numerically, server-side) and asks the configured LLM for a concise business summary + action suggestions. Returns { ringkasan }.

Shared Helpers

Two utilities worth knowing when reading the source:

```
// Reads settings/komisi — perLayanan[serviceId] override, else global, clamped 0-100.
async function komisiPersenUntuk(serviceId: string): Promise<number>

// Recomputes services.jumlahKru = count of active kru (status undefined/null/'aktif')
// whose keahlian is empty (generalist) or includes the serviceId. Drives the
// capacity-based booking slot mechanism.
async function recomputeKapasitasLayanan(): Promise<void>
```

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Security Rules

firebase/firestore.rules — the second line of defense alongside Cloud Functions.
Read/write access per collection.

Helper Functions

```
function isSignedIn() { return request.auth != null; }
function role() { return get(/databases/$(db)/documents/users/$(request.auth.uid)).data.role; }
function isOwner(uid) { return isSignedIn() && request.auth.uid == uid; }
function isAdmin() { return isSignedIn() && role() == 'admin'; }
```

Access Control Matrix

Collection	Read	Create	Update	Delete
`users/{userId}`	owner or admin	owner only	owner or admin	admin
`services`	public	—	admin	admin
`slots/{slotId}`	any signed-in	signed-in, terisi must be exactly +1	same rule, or admin	admin
`orders/{orderId}`	participant or admin	signed-in owner; price fields must match services	admin, or participant limited to status/photos/cancellation	admin

Collection	Read	Create	Update	Delete
`orders/{id}/messages`	participant or admin	participant or admin	—	—
`kru/{cleanerId}`	public	admin	owner/admin, or signed-in diff-only on rating fields	admin
`settings/ai`	admin only	admin	admin	admin
`settings/{other}`	signed-in, or public if docId=='app'	admin	admin	admin
`payments`	owner or admin	signed-in owner	admin only	—
`reviews`	public	signed-in owner	admin	admin
`notifications`	owner	any signed-in (ideally server-written)	—	—
`vouchers/{kode}`	signed-in	admin	admin, or diff-only terpakai == old+1	admin
`voucherUsages`	get: any signed-in; list: admin/owner	signed-in owner	admin	admin
`broadcasts / banners`	admin / public	admin	admin	admin
`admin2fa/{uid}`	owner	owner	owner	owner
`payouts`	admin or owning cleaner	admin or signed-in (in completion tx)	admin	admin
`kasKru`	admin or owning cleaner	—	admin only	admin
`setoran`	admin or owning cleaner	admin	admin	admin
`referralClaims`	admin or any signed-in	—	admin only	admin

Collection	Read	Create	Update	Delete
`disputes`	admin or the filer	signed-in filer, status must be diajukan	admin	admin

Key Invariants

> ⚠ Order money/ownership fields are immutable to non-admins once created — a client can only ever touch `['status', 'fotoSebelum', 'fotoSesudah', 'penugasan']` on an existing order, and cancellation is only permitted from pre-assignment statuses.

Other write-shape constraints enforced at the rules layer:

- Slot `terisi` can only be incremented by exactly +1 per client write — decrements are admin-SDK only (the atomic-locking mechanism).
- Voucher `terpakai` can only increment by exactly 1 per write.
- `kru` rating fields (`rataRating`, `jumlahUlasan`) can only move via a controlled delta matching a genuine review submission.
- `kasKru` balances are never client-writable — only Cloud Functions (admin SDK, which bypasses rules entirely) or admin-authenticated writes can change them.

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Commission & Payouts

BUSINESS LOGIC

Commission & Payouts

The exact algorithms behind money in Tuntaskilat — every formula here is enforced server-side, never trusted from the client.

Wage Splitting — bagiUpah()

`packages/tk_core/lib/models/wage.dart` — splits a completed order's total between platform commission and crew, weighted by role.

```
totalInt   = round(total)
komisi     = round(totalInt × komisiPersen / 100)     // platform commission
pool       = totalInt - komisi                        // remaining for crew
totalBobot = Σ peran.bobot for each crew member         // worker=1.0, helper=0.6

for each crew member p:
    bagian[p] = floor(pool × p.peran.bobot / totalBobot)

sisa = pool - Σ bagian                                  // rounding remainder
bagian[crew.first] += sisa                               // remainder → lead/first member

# Guarantee: komisi + Σ bagian == total, exactly. No Rupiah lost or created.
```

⚠ The server-side `onOrderFinalize` Cloud Function reimplements the same weighted split inline, but rounds each share *individually* rather than floor + remainder-to-first — so its per-share numbers can differ from the client's `bagiUpah()` preview by ±1 for the same input. The client value is only a preview; the Cloud Function is authoritative for actual payouts.

Commission Config — KonfigKomisi

```
persenUntuk(serviceId) = clamp(perLayanan[serviceId] ?? komisiPersen, 0, 100)
```

Mirrored server-side as `komisiPersenUntuk(serviceId)`, reading `settings/komisi`. Global default is 20%.

Cash Deposit Ledger — KasKru

For cash-paid orders, the crew member collects the full amount from the customer in person — but the platform's commission share is still owed. That amount is tracked as a debt (`saldoTunai`) the crew must periodically settle with the office.

```
saldoTunai == totalMasuk - totalSetor        // audit invariant
bolehTerimaTunai == saldoTunai ⩽ batasNunggak   // default batasNunggak = Rp200,000
```

`saldoTunai` increases exactly once per cash order, booked by `onOrderFinalize` (guarded by the `kasTunaiDibukukan` flag) — no client path can write it directly (Security Rules restrict `kasKru` writes to admin). It decreases only when an admin records a deposit via `terimaSetoran()`, which runs inside a transaction that rejects any deposit amount exceeding the current balance.

Voucher Discount Formula

```
if !aktif                          → reject 'nonaktif'
if now > berlakuHingga             → reject 'kadaluarsa'
if sisaKuota ⩽ 0                   → reject 'kuotaHabis'   // sisaKuota = kuota⩽0 ? ∞ : kuota-terpakai
if subtotal < minBelanja           → reject 'minimalBelanja'

potongan = tipe=='persen' ? subtotal × nilai / 100 : nilai
if tipe=='persen' && maxPotongan>0 && potongan>maxPotongan
    potongan = maxPotongan
if potongan > subtotal
    potongan = subtotal
```

The server-side `potonganFormulaVoucher()` applies the same nominal-cap math WITHOUT the kuota/expiry/aktif gates (those are enforced client-side inside the booking transaction) — its purpose is purely to cap the claimed amount: `potonganBenar = min(potonganKlien, formulaMax)`, preventing a client from claiming more discount than the voucher's own math allows.

Voucher Anti-Abuse

Two independent guards inside `buatPesananLengkap()`:

New-user-only vouchers

Checked *outside* the transaction (a query for any prior order by `userId`) — if the user has ordered before, throws `VoucherException(hanyaPenggunaBaru)`.

Per-phone-number lock

Doc ID `voucherUsages/{kode}__{normalizedPhone}` (phone normalized: `0xxxx → 62xxxx`, non-digits stripped). Inside the transaction, if that doc already exists → reject. This is keyed on **phone number, not uid** — specifically to prevent the "make a new account, reuse the phone number" abuse vector. `voucher.terpakai` is incremented by exactly 1 in the same transaction, matching the Security Rules constraint.

Capacity-Based Slots

Replaces a legacy boolean "taken" lock with a live counter. `services.jumlahKru` — maintained by `recomputeKapasitasLayanan()`, triggered by `onKruDitulis` / `onLayananDibuat` / `backfillKapasitasKru` — is the count of active crew eligible for that service (empty `keahlian` = generalist, or `keahlian` includes the serviceId). This becomes the hourly booking capacity.

```
kapasitas = services.jumlahKru  (fallback 1 if field absent)
terisi ≥ kapasitas   → JadwalPenuhException
kapasitas ≤ 0        → TidakAdaKruException
```

A service with zero eligible crew is fully locked for booking. Cancellation (`onOrderCancelled`) decrements `terisi` via the admin SDK — the only permitted decrement path.

Cash Arrears Guard

```
if (order.tunai && !await bolehTerimaTunai(lead.cleanerId)) {
  throw KruNunggakException(lead.nama);
}
```

Before assigning a cash-payment order to a lead crew member, `tugaskanKruMulti()` checks `kasKru/{lead}.bolehTerimaTunai`. A crew member who has accumulated more than the arrears threshold in un-deposited cash commission cannot be assigned new cash-paying jobs until they settle up.

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Security Rules

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Payments & Gateway

INTEGRATIONS

Payments & Gateway

Payment methods are pluggable — a manual verification flow ships by default, with Xendit available as an opt-in dynamic gateway.

Gateway Abstraction

`packages/tk_core/lib/services/payment_gateway.dart` defines a `PaymentGateway` interface to avoid vendor lock-in:

```
abstract class PaymentGateway {
  Future<InstruksiBayar> buatInstruksi({
    required String orderId,
    required int jumlah,
    required MetodeBayar metode,
  });
  String get nama;
}

class StaticGateway implements PaymentGateway { ... }  // active by default
class XenditGateway implements PaymentGateway { ... }   // opt-in
```

Static vs. Dynamic

	StaticGateway	XenditGateway
Instructions	Fixed bank account / QRIS image from settings/pembayaran	Per-order dynamic VA / QRIS from Xendit
Verification	Manual — customer uploads proof, admin approves/rejects	Automatic — settled via webhook
Requires	Nothing extra (default MVP flow)	XENDIT_SECRET_KEY + XENDIT_CALLBACK_TOKEN env vars
Status if unconfigured	Always available	buatTagihanXendit throws failed-precondition; xenditWebhook returns HTTP 501

Both paths land on the same `payments` collection and, once `terverifikasi`, feed into the same commission / payout calculation — they're layered, not competing systems.

Xendit Flow

```
1. Customer selects a Xendit-backed method at checkout
2. App calls buatTagihanXendit({ orderId, amount })
     → verifies caller owns the order
     → POST https://api.xendit.co/v2/invoices  (Basic auth from XENDIT_SECRET_KEY)
     → persists { gateway, invoiceId, invoiceUrl } on the payment doc
     → returns invoiceUrl for the app to open
3. Customer completes payment on Xendit's hosted page
4. Xendit calls xenditWebhook with x-callback-token + { external_id: orderId, status }
     → validates token (401 if wrong)
     → on PAID/SETTLED: payments.statusBayar = 'terverifikasi', orders.status = 'terverifikasi'
```


Commission vs. Payment Config — Two Different Things

It's easy to confuse `settings/komisi` and `settings/pembayaran` because both live under "money settings" in the admin panel — but they govern orthogonal concerns:

Setting	Controls
`settings/komisi`	The platform's revenue-share percentage taken from each completed order, used by bagiUpah/onOrderFinalize to split money between platform and crew.
`settings/pembayaran`	Which customer-facing payment methods are offered and whether each is statis (manual) or dinamis (Xendit).

A `dinamis` payment method still goes through the same commission calculation once the order completes — the two systems are layered, not competing.

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Commission & Payouts

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AI Features

INTEGRATIONS

AI Features

Two AI-powered callables — a customer-facing support chat and an admin-facing business analyst — both provider-agnostic.

Multi-Provider Config

`settings/ai` (admin-only, read/write) selects which provider is active and holds its key:

Field	Notes
provider	gemini | openai | anthropic
geminiApiKey / openaiApiKey / anthropicApiKey	per-provider key field; falls back to env vars (GEMINI_API_KEY, etc.) if empty
model	defaults: gemini-2.0-flash, gpt-4o-mini, claude-haiku-4-5
aktif	master on/off switch

⚠ Because keys fall back to environment variables, a deployed env var can silently supply a key even if the admin panel shows none configured — worth checking both when debugging "AI not responding".

A single dispatcher, `panggilAi()`, routes to the correct REST endpoint per provider: Gemini's `generateContent` (key in query string), OpenAI's `/chat/completions`, or Anthropic's `/v1/messages`.

Customer Service AI — csAi

```
// onCall, auth required
Input:  { pertanyaan: string, orderId?: string }
Output: { jawaban: string }
```

The system prompt is grounded in the **live active-service catalog and prices**, fixed operating hours (08.00–19.00 WIB), the 3 payment methods, and cancellation policy. If `orderId` is supplied and belongs to the calling uid, that order's current status is injected as additional context — the assistant only ever sees the caller's own data.

Fallback: if the AI config's key is missing or `aktif` is false, the function throws `failed-precondition` ("Asisten AI belum diaktifkan").

Business Analyst AI — analisaBisnisAi

```
// onCall, admin-only
Output: { ringkasan: string }
```

Purely numeric aggregation is done server-side in TypeScript — the LLM never touches the database directly. Over the trailing 30 days of `orders`, the function computes: total / selesai / dibatalkan counts, omzet (revenue from completed orders), and a per-service order-count breakdown — then hands those numbers to `panggilAi()` with a prompt capped at 6 bullet points plus 2–3 concrete action suggestions.

Guardrails

- The prompt explicitly forbids fabricating data or leaking other customers' PII.
- `analisaBisnisAi` is told not to fabricate beyond the numbers it was given.
- Both functions require the caller to be authenticated; the business analyst additionally requires `role == 'admin'`.

INTEGRATIONS

Notifications

Two independent notification channels — Firebase Cloud Messaging for push, and WhatsApp via Fonnte for a persistent paper trail.

FCM Push Events

Function	Recipient	Trigger
onOrderAssigned	assigned crew	order enters 'ditugaskan'
pushStatusPelanggan	customer	terverifikasi / ditugaskan / dalam_perjalanan / dikerjakan / selesai / ditolak
onChatMessageCreate	the other chat participant	any new message in orders/{id}/messages
reminderKru	assigned crew	every 30 min, if jadwal is 90–150 min away
onBroadcastCreate	all pelanggan with a token	admin creates a broadcasts/{id} doc

ⓘ Invalid/expired tokens are cleaned from `kru.fcmTokens` automatically whenever a push fails with `registration-token-not-registered` or `invalid-argument`.

WhatsApp (Fonnte) — waNotifOrder

A thin abstraction, `WaSender`, with `FonnteSender` as the active implementation — swap the class if you switch WhatsApp providers without touching call sites. Fires on the same order-status transitions as the push notification (terverifikasi / ditugaskan / selesai), sending a formatted text message to the customer's phone. No-op — safely, silently — if `FONNTE_TOKEN` is unset in `firebase/functions/.env`.

In-App Notification Feed

Every push additionally writes a `notifications` document (see Data Model) so the customer/crew has a persistent, scrollable history inside the app — not just a transient system notification. Notifications are deep-linkable via the `orderId` field.

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AI Features

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Deployment

OPERATIONS

Deployment

One Firebase project, multiple hosting sites, one Cloud Functions codebase.

Hosting Targets

`firebase.json` declares multiple hosting entries, each mapped to a distinct Firebase Hosting site via `.firebaserc` target aliases:

Target	Public dir	Serves
admin	apps/admin/build/web	Admin panel (Flutter web build)
manualbook	docs-site/manualbook/dist	This end-user manual
documentation	docs-site/documentation/dist	This developer reference

Site IDs must be globally unique across all Firebase projects. If a bare name like `manualbook` is taken, Firebase assigns a project-prefixed fallback (e.g. `tuntaskilat-manualbook`) — check the actual `*.web.app` URL printed after `firebase hosting:sites:create`.

Cloud Functions Env Vars

Optional integrations are gated behind environment variables in `firebase/functions/.env` — every one of them degrades gracefully (no-op, HTTP 501, or a clear `failed-precondition` error) when unset, so partial configuration never breaks the core booking flow.

```
# WhatsApp notifications (waNotifOrder) — no-op if unset
FONNTE_TOKEN=

# Xendit dynamic payments — buatTagihanXendit throws if unset
XENDIT_SECRET_KEY=

# Xendit webhook — xenditWebhook returns HTTP 501 if unset
XENDIT_CALLBACK_TOKEN=

# AI fallback keys (settings/ai Firestore fields take precedence)
GEMINI_API_KEY=
OPENAI_API_KEY=
ANTHROPIC_API_KEY=
```

Deploy Commands

```
# Build the docs sites
cd docs-site/manualbook && npm run build && cd ../..
cd docs-site/documentation && npm run build && cd ../..

# Build the admin web app
cd apps/admin && flutter build web --release && cd ../..

# Deploy everything
firebase deploy --only functions,firestore:rules,storage:rules,hosting

# Or scope to just the docs sites
firebase deploy --only hosting:manualbook,hosting:documentation
```

Pre-Deploy Checklist

- `flutter analyze` clean across all 4 packages (tk_core, admin, pelanggan, kru).
- `flutter test packages/tk_core apps/admin` — all green.
- `cd firebase/functions && npx tsc -p .` — no type errors.
- Firestore Security Rules reviewed for any collection touched this release.
- New Cloud Functions match the region convention (`asia-southeast2`).

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Notifications

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Changelog & Roadmap

OPERATIONS

Changelog & Roadmap

Development phases, from the initial thesis-scoped MVP through the current business-layer feature set.

FASE 1

Bug Blockers

- Splash/launcher fixes
- P6 billing blank-screen fix
- Voucher rules deploy
- Multi-crew assignment wiring
- Profile photos
- Payment proof + QRIS display
- Dashboard status chart
- System bar consistency audit

FASE 2

Customer UX

- Wizard-style booking form
- Saved addresses + map picker

- Reverse-geocoding (Nominatim)
- Full catalog with category filter
- Active-order home card
- Cancel + CS contact from tracking
- Granular permissions + Terms
- Chat + call wiring

FASE 3

Crew (Kru)

- FCM push + assignment reminders (cron)
- Real-road OSRM routing
- Custom notification sound

FASE 4

Admin

- Full dynamic pricing catalog editor
- Crew lifecycle (keahlian/tipe/status)
- Skill-based assignment matching
- New-user + per-phone voucher anti-abuse
- Service area module
- TOTP 2FA
- Admin team management (buatAdmin/setNonaktifAdmin)

FASE 5

Business Systems

- Commission config (global + per-service)
- Cash deposit ledger (kasKru/setoran)
- Referral program with anti-abuse
- Modular payment gateway (Static + Xendit stub)
- WhatsApp notifications (Fonnte)

FASE 6

Quality & Release

- Maintenance mode / forced update (settings/app)
- Skeleton loading states
- Memory-leak audit (chat screen controllers)
- Extended black-box test coverage

POST-FASE 6

Business Expansion

- Multi-provider AI (Gemini/OpenAI/Anthropic) — CS chat (csAi) + business analyst (analisaBisnisAi)
- Xendit dynamic invoicing activated (buatTagihanXendit + xenditWebhook)
- Capacity-based booking (replaces boolean slot lock with a live crew-count counter)
- AI0 Pantau Operasional — live crew map + chat quality monitoring

- A9 Kelola Klien — customer roster with spend aggregation
- Broadcast push notifications + banner promo content
- Auto-cancel unassigned orders after 24h
- Server-authoritative review rating aggregation

For the exact commit history, see the repository's Git log — each feature phase was developed on its own branch and merged via pull request into `dev`.